Zixiang Xu

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Ph.D. Student
Computer Science
University of Southern California
Los Angeles, CA

Education

University of Southern California (USC) *Los Angeles, CA*
Ph.D. Student in Computer Science · Advisor: Prof. Yue Zhao Aug. 2026 – Present
FORTIS Lab. Expected graduation: 2030.

Research Experience

Meta AI *Remote*
Research Collaboration Sep. 2026 – Present
Lead Grid-Bench: executable spatial environments for frontier-model evaluation and post-training.

Tencent (WeChat AI) *Beijing, China*
Research Intern · Foundation Model Group Jun. 2026 – Aug. 2026
Contributed to WeLM post-training. Led AlgoWorlds, building 240 verifiable tool-use worlds across ten optimization families and generating agent trajectories for post-training.

Alibaba Group *Beijing, China*
Research Intern Mar. 2026 – May 2026
Led Inside the Unfair Judge, studying scoring-bias mechanisms, causal bias control, and failure prediction to inform the reliability of automated feedback for post-training.

MBZUAI *Abu Dhabi, UAE*
Research Intern · Supervisor: Prof. Xiuying Chen Mar. 2025 – Aug. 2025
Led SocialMaze, GTA, and DyFlow: social-reasoning post-training with curated traces, algorithmic graph reasoning, and dynamic agent workflows.

MINE Lab, University of Notre Dame *Remote*
Research Intern · Supervisor: Prof. Xiangliang Zhang Oct. 2024 – Aug. 2025
Led Cross-Lingual Pitfalls and Adaptive Distraction, automating failure discovery and using cross-lingual failures for targeted post-training and transfer evaluation.

MIT Center for Transportation & Logistics *Massachusetts, USA*
Summer Research Student · Supervisor: Prof. Elenna R. Dugundji Aug. 2024
Led Temporal-IRL, applying inverse reinforcement learning to port congestion and berth scheduling; selected for a TRB 2025 oral presentation.

Microsoft (Asia) Internet Engineering Institute *Beijing, China*
Research Intern · Supervisor: Wang Pan Jan. 2024 – Mar. 2024
Led LLM and VLM evaluation, including GPT-4V capability assessment and reproduction.

Honors and Awards

USC Annenberg Fellowship 2026
Competitive fellowship top-off for selected USC Viterbi Ph.D. students.

CCF Elite Collegiate Award 2024
One of 100 junior and senior undergraduates selected nationwide; recognized for academics, research, and service. Invited speaker/session host at CNCC; team placed **3rd nationwide** in the CCSP Team Contest.

Gold Medal, The 2022 ICPC Asia Jinan Regional Contest 2022
Gold medal in the ICPC, the world's premier collegiate algorithmic programming competition.

Silver Medal, National Olympiad in Informatics (NOI) 2020
Silver medal in China's national informatics olympiad for secondary-school students, organized by the China Computer Federation (CCF).

Tools and Platforms

RewardAudit. Model-free audits of reward functions and verifiers for RLVR; open-source release forthcoming.
MathInsight Arena. Expert, blind comparisons of how AI poses, evaluates, and develops mathematical research problems.

Publications

[1] Grid-Bench: Can LLMs Predict How Discrete Worlds Change?
Zixiang Xu, Zirui Song, Wanrong Zheng, Chiyu Zhang, Yue Huang, Yuyang Peng, Kexin Zhang, Jinkun Chen, Xiuying Chen, Xiangliang Zhang, Yan Liu, Laurent Itti, Frank Shyu, Xiaoqiang Luo, Sandeep Pandey, Xi Liu, Luke Simon, Xiyang Hu, Yue Zhao.
Preprint, 2026. [Project] [Paper] [Code] [Data]

[2] AlgoWorlds: Benchmarking Tool Use for Global Optimization in Algorithmic Worlds
Zixiang Xu, Jiaan Wang, Fandong Meng.
Preprint, 2026. [Project] [Paper] [Code & Data]

[3] agent-risk: Risk Assessment for Changes to AI Agents
Yue Zhao, **Zixiang Xu**, Angela Huyue Zhang, S. Alex Yang, Xiyang Hu.
Preprint, 2026. [Paper] [Code & Data]

[4] SocialMaze: A Benchmark for Evaluating and Enhancing Social Reasoning in Large Language Models in Complex Social Environments
Zixiang Xu, Yanbo Wang, Yue Huang, Haomin Zhuang, Yujun Zhou, Jiayi Ye, et al.
Findings of EMNLP 2026; previously SocialSim @ **COLM 2025** Workshop (Spotlight).
[Project] [Paper] [Code] [Data]

[5] Cross-Lingual Pitfalls: Automatic Probing Cross-Lingual Weakness of Multilingual Large Language Models
Zixiang Xu[*], Yanbo Wang[*], Yue Huang[*], Xiuying Chen, Jieyu Zhao, Meng Jiang, Xiangliang Zhang ([*]Equal contribution).
ACL 2025. [Project] [Paper] [Code]

[6] Adaptive Distraction: Probing LLM Contextual Robustness with Automated Tree Search
Yanbo Wang[*], **Zixiang Xu**[*], Yue Huang[*], Chujie Gao, Siyuan Wu, Jiayi Ye, et al. ([*]Equal contribution).
NeurIPS 2025. [Paper] [Code]

[7] DyFlow: Dynamic Workflow Framework for Agentic Reasoning
Yanbo Wang, **Zixiang Xu**, Yue Huang, Xiangqi Wang, Zirui Song, Lang Gao, Chenxi Wang, et al.
NeurIPS 2025. [Paper] [Code]

[8] JevOut: Natural Context Can Flip Decision Models
Zixiang Xu, Zirui Song, Chiyu Zhang, Xiuying Chen, Xi Liu, Xiyang Hu, Yue Zhao.
Preprint, 2026. **Agenthon @ NeurIPS 2026** (Poster).
[Project] [Paper] [Code & Data]

[9] Inside the Unfair Judge: A Mechanistic Interpretability Account of LLM-as-Judge Bias
Zixiang Xu, Sixian Li, Huaxing Liu, Xiang Wang, Shuai Li, Zirui Song, Xiuying Chen.
Preprint, 2026. [Project] [Paper]

[10] GTA: Graph Theory Agent and Benchmark for Algorithmic Graph Reasoning with LLMs
Zixiang Xu, Yanbo Wang, Chenxi Wang, Lang Gao, Zirui Song, Yue Huang, Zhaorun Chen, Xiangliang Zhang, Xiuying Chen.
Preprint, 2026. [Project] [Paper] [Code & Data]

Publications (continued)

[11] Do LLMs "Feel"? Emotion Circuits Discovery and Control
Chenxi Wang, Yixuan Zhang, et al., **Zixiang Xu**, et al.
ICML 2026. [Paper] [Code]

[12] Audio Jailbreak: An Open Comprehensive Benchmark for Jailbreaking Large Audio-Language Models
Zirui Song, Qian Jiang, et al., **Zixiang Xu**, et al.
ACL 2026. [Paper] [Code]

[13] ManipLVM-R1: Reinforcement Learning for Reasoning in Embodied Manipulation with Large Vision-Language Models
Zirui Song, Guangxian Ouyang, et al., **Zixiang Xu**, et al.
AAAI 2026. [Paper]

[14] Temporal-IRL: Modeling Port Congestion and Berth Scheduling with Inverse Reinforcement Learning
Yikuan Hu*, **Zixiang Xu***, Wei Zhang*, et al. (*Equal contribution).
TRB 2025 (Oral). [Paper]

[15] Under the Shadow of Babel: How Language Shapes Reasoning in LLMs
Chenxi Wang, Yixuan Zhang, Lang Gao, **Zixiang Xu**, et al.
Findings of EMNLP 2025. [Paper]

[16] GenUDC: High-Quality 3D Mesh Generation with Unsigned Dual Contouring Representation
Ruowei Wang, Jiaqi Li, et al., **Zixiang Xu**, et al.
ACM MM 2024. [Paper]

[17] Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies
Zirui Song, Yuan Huang, Junchang Liu, et al., **Zixiang Xu**, et al.
EMNLP 2026. [Paper]

[18] Evaluate Bias without Manual Test Sets: A Concept Representation Perspective for LLMs
Lang Gao, Kaiyang Wan, Wei Liu, et al., **Zixiang Xu**, et al.
Preprint, 2025. [Paper]

[19] MTFusion: Reconstructing Any 3D Object from Single Image Using Multi-word Textual Inversion
Yu Liu, Ruowei Wang, Jiaqi Li, **Zixiang Xu**, Qijun Zhao.
PRCV 2024 (Oral). [Paper]

[20] Text-Based Face Retrieval: Methods and Challenges
Yuchuan Deng, Qijun Zhao, Zhanpeng Hu, **Zixiang Xu**.
CCBR 2023 (Oral). [Paper]

Reviewing Service

Conference Reviewer: AAAI 2027; ICLR 2026–2027; COLM 2026; NeurIPS 2026; NLPCC 2025.

ACL Rolling Review Reviewer: February and May 2025; August and October 2026 cycles.

Workshop Reviewer: VLM4RWD at NeurIPS 2026; GroundLM and REALM at EMNLP 2026; Social Simulation with LLMs at COLM 2025.